An

Industrial Training Report

on

ShopEase: Full-Stack E-Commerce Web Application Using MERN Stack

submitted
in partial fulfillment
for the award of the Degree of
Bachelor of Technology

Department of Computer Science & Engineering



Submitted to:

Neha Mathur

Designation: Assistant Professor

Submitted by:

Name of Student: Anshu Verma

Roll No: 24ESKCA025

DECLARATION

I hereby declare that the Industrial Training Report entitled **"ShopEase: Full-Stack E-Commerce Web Application Using MERN Stack"** is a structured record of the work and learning undertaken during my industrial training. The report has been prepared for the fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering at Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur.

The project described in this report is presented as a training/prototype implementation developed to demonstrate MERN stack concepts including React-based user interfaces, Node.js and Express REST APIs, MongoDB data management, authentication, CRUD operations and responsive web design.

To the best of my knowledge, this report has not been submitted elsewhere for any other academic qualification.

(Signature of Student)

Anshu Verma
24ESKCA025

Date: _____

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TO
Anshu Verma

Weare please tooffer you the position of Mern Stack Deveoper at Ice Hut Technologies. This offer is based on our discussions and your qualifications, and we are confident that your experience will be a valuable asset to our organization.

- Position: Mern stack Developer
- Start Date : 16-05-2026 End Date :30-06-2026
- Mode: Offline

Working Schedule:

- Working Days: Monday to Saturday
- Training Timings: 9:00 AM to 10:00 AM
- Internship Working Timings: 10:00 AM to 6:00 PM
- Lunch Break: 1:00 PM to 1:45 PM

Roles & Responsibilities:

- Develop user interfaces using React.js
- Create reusable UI components
- Ensure responsive design (mobile + desktop)
- Handle state management (Redux / Context API)
- Optimize performance and user experience

Stipend / Compensation:

- Please note that this is an unpaid position, and no stipend or monetary compensation will be offered for the duration of this role

Terms & Conditions:

- Internship does not guarantee permanent employment
- Performance will be evaluated periodically
- Confidentiality must be maintained for all projects and data

If you accept this offer, please sign below as confirmation. We look forward to supporting your career growth.

Best regards,

General Manager

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CERTIFICATE



The above certificate was issued by Ice Hut Technologies Pvt. Ltd. for Anshu Verma's MERN Stack training. It records the offline training period from 16 May 2026 to 30 June 2026 under the guidance of Mr. Gajendra Badra. ■

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Ice Hut Technologies Pvt. Ltd.** for providing me the opportunity to undergo industrial training in MERN Stack Development. The training helped me understand how concepts learned in the classroom are applied while building practical web applications.

I am thankful to **Mr. Gajendra Badra**, Project Coordinator, for his guidance and technical direction during the training. His support helped me understand frontend development with React.js, reusable components, responsive interfaces and the overall workflow of a modern full-stack application.

I also express my sincere thanks to **Mrs. Neha Mathur**, Assistant Professor, Department of Computer Science and Engineering, SKIT, for her academic guidance, motivation and support in completing this report.

Finally, I thank my teachers, classmates and everyone who directly or indirectly contributed to the successful completion of my industrial training and this report.

(Signature of Student)

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ABSTRACT

The rapid growth of online commerce has increased the need for responsive, secure and scalable web applications that can manage products, customers, carts and orders through a single platform. This report presents **ShopEase**, a MERN Stack based e-commerce web application developed as a training/prototype project to demonstrate full-stack web development concepts.

The application uses **React.js** for the frontend, **Node.js** with **Express.js** for REST API development and **MongoDB** for persistent data storage. The system is organised around reusable UI components and API-driven communication. Major modules include user registration and login, product browsing, product management, shopping cart, order creation and an administrative dashboard.

The training also focused on responsive design, state management, form validation, CRUD operations, API integration, error handling and basic application security. The prototype demonstrates how a complete web solution can be developed by integrating frontend, backend and database layers into a single application.

This report documents the project objectives, technology stack, system architecture, implementation workflow, interface snapshots, testing observations, results, limitations and future enhancements.

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Chapter 1
Introduction

1.1 Project Description

ShopEase is a training-oriented full-stack e-commerce web application designed to demonstrate the complete MERN development lifecycle. The application allows a customer to browse products, view product details, add items to a shopping cart and place an order. An administrative interface is included for managing products and viewing order information.

The project follows a client–server architecture. React renders the user interface and communicates with the backend through REST APIs. Express.js handles routing and business logic, while MongoDB stores users, products, carts and orders. This separation makes the application modular and easier to maintain.

1.2 Project Scope

The scope covers user authentication, product CRUD operations, product listing and filtering, cart management, order creation, admin dashboard, responsive design and REST API integration. The implementation is intended for learning and demonstration rather than production-scale commerce.

1.3 Objectives

- Understand the architecture of a modern MERN stack application.
- Develop responsive interfaces using React and reusable components.
- Build REST APIs using Node.js and Express.js.
- Store and retrieve application data using MongoDB.
- Implement authentication, validation, CRUD operations and error handling.
- Understand integration and testing of frontend and backend modules.

1.4 Project Features

Key features include registration/login, product catalogue, search and category filtering, shopping cart, order placement, admin product management, dashboard statistics, responsive layouts and API-based data exchange.

Chapter 2
Tools & Technology Used

2.1 Software Model Used

An iterative development model was followed. The application was developed in small stages: UI structure, product module, backend APIs, database integration, authentication, cart and order workflows, followed by testing and interface refinement. This approach allowed individual modules to be tested before complete integration.

2.2 Programming Languages, Libraries & Frameworks

Component	Technology
Frontend	React.js, HTML5, CSS3, JavaScript
Backend	Node.js, Express.js
Database	MongoDB
API communication	REST API, JSON, Axios
Development tools	VS Code, npm, Git/GitHub
UI approach	Responsive CSS / reusable React components

Table 2.1: Technology stack used in the project

2.3 Major Application Modules

Module	Purpose
Authentication	Registration, login and protected user/admin actions.
Product Management	Create, read, update and delete product records.
Product Catalogue	Display products with category/search based filtering.
Shopping Cart	Add, update and remove products before checkout.
Order Management	Create orders and display order status/details.
Admin Dashboard	Summarise products and orders for administration.

Table 2.2: Major application modules

2.4 System Architecture

The system follows a three-layer structure. The React client handles presentation and user interactions. The Express/Node.js server exposes REST endpoints and contains business logic. MongoDB provides persistent storage. Requests flow from the browser to the API and back as JSON responses.



Figure 2.1: System architecture and administrative view of ShopEase

Chapter 3
Snapshots / Proposed Work

3.1 Processing Flow

The complete processing flow begins when a user opens the application. The frontend sends authentication and product requests to the Express server. The server validates the request, performs business logic and communicates with MongoDB. The response is returned as JSON and React updates the interface.

1	User opens ShopEase
↓	
2	React interface sends API request
↓	
3	Express/Node.js validates request
↓	
4	MongoDB reads or updates data
↓	
5	JSON response returned to React
↓	
6	Updated page / cart / order displayed

Figure 3.2: End-to-end processing flow

3.2 Database Design

MongoDB is used because its document-oriented model is convenient for storing product, user and order information. Typical collections are users, products, carts and orders. References between documents are maintained using identifiers so that order and cart records can be associated with the appropriate user and products.

Collection	Important fields	Purpose
users	_id, name, email, passwordHash, role	Stores customer/admin accounts.
products	_id, name, category, price, stock, image	Stores product catalogue.
carts	userId, items[], total	Stores current cart items.
orders	userId, items[], total, status, createdAt	Stores placed orders.

Table 3.1: MongoDB collections and important fields

3.3 Project Screenshots

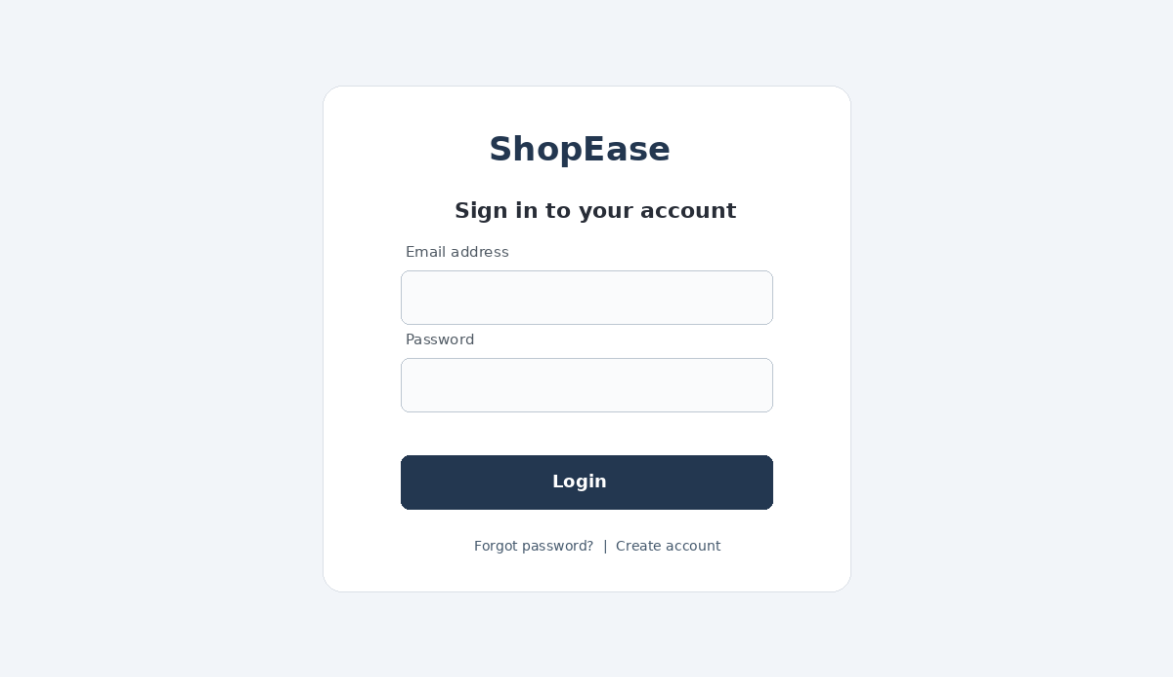


Figure 3.3: ShopEase login interface (illustrative training-project screen)

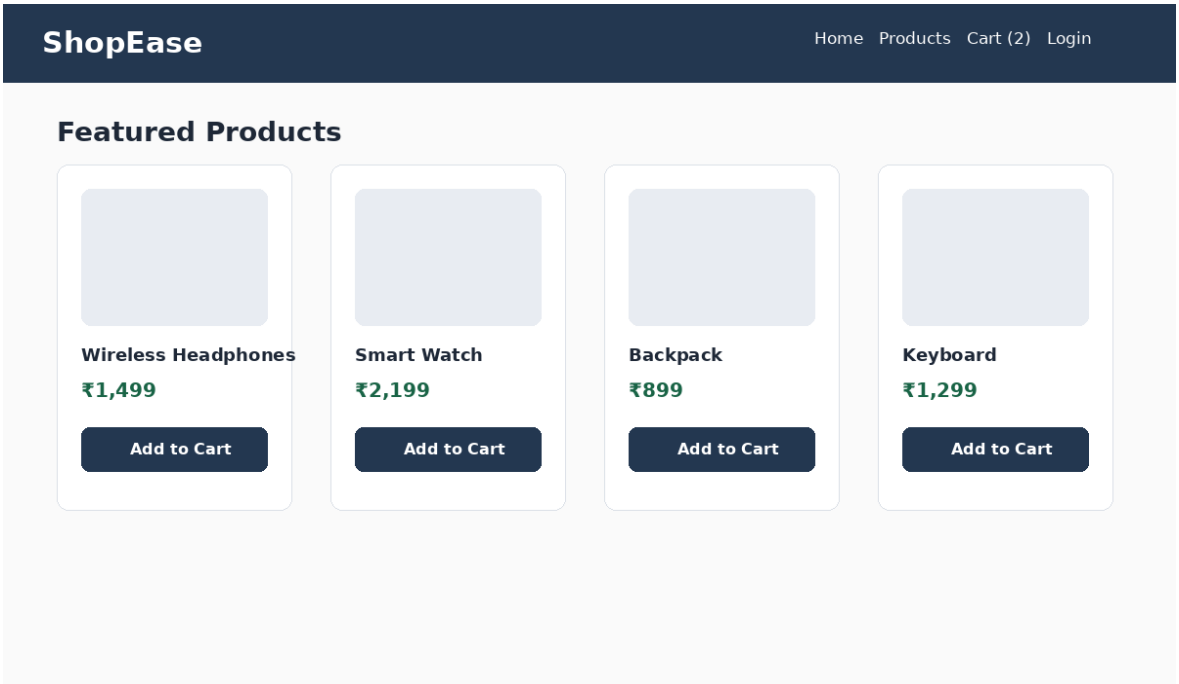


Figure 3.4: Product browsing interface (illustrative training-project screen)



Figure 3.5: Administrative dashboard (illustrative training-project screen)

3.4 Code Snippets

Express route example

```
app.get("/api/products", async (req, res) => {
  const products = await Product.find();
  res.json(products);
});
```

React data loading example

```
useEffect(() => {
  axios.get("/api/products").then(res => setProducts(res.data));
}, []);
```

These snippets represent the type of frontend-backend integration demonstrated during the training project.

Chapter 4
Results and Discussion

4.1 Functional Testing

Functional testing was performed module by module. Each major user action was checked for expected behaviour, validation and correct API/database interaction.

Test Case	Expected Result	Status
User registration	Valid user record created	Pass
User login	Authenticated session/access granted	Pass
Product listing	Products displayed from database	Pass
Search/filter	Relevant products displayed	Pass
Add to cart	Selected product added to cart	Pass
Update cart quantity	Cart total recalculated	Pass
Place order	Order record created	Pass
Admin product update	Product changes reflected in catalogue	Pass
Responsive layout	Interface usable on common screen sizes	Pass

Table 4.1: Functional testing results

4.2 Results

The training project successfully demonstrates integration of the three MERN layers. Product information can be represented as MongoDB documents, retrieved through Express REST endpoints and rendered through reusable React components. The modular design also makes it possible to extend the system with payment processing, advanced search and cloud deployment.

The main learning outcome was an understanding of how frontend components, backend routes and database operations are connected in a complete application rather than being developed as isolated pieces.

4.3 Discussion

The project showed that a clean separation between presentation, API logic and persistence simplifies debugging and maintenance. Reusable components reduce duplication in the frontend, while REST endpoints provide a predictable interface between the client and server. Input validation and error handling remain important because incorrect data can otherwise propagate into the database.

The prototype should be treated as a training implementation. Production deployment would require stronger security controls, proper password hashing and token management, rate limiting, payment security, logging, automated tests and cloud infrastructure.

Chapter 5
Conclusion and Future Scope

5.1 Conclusion

The industrial training provided practical exposure to MERN Stack Development through the design of ShopEase, a full-stack e-commerce training prototype. The project brought together React.js, Node.js, Express.js and MongoDB to implement a complete application flow from user interaction to database persistence.

The work strengthened understanding of component-based UI development, REST API design, CRUD operations, database modelling, responsive layouts and frontend-backend integration. It also provided experience in breaking a larger application into manageable modules and testing each part before integration.

5.2 Limitations

- The project is a training/prototype implementation and is not intended to process real financial transactions.
- Payment gateway integration is not included in the prototype.
- Advanced production-grade authentication and authorisation controls require further implementation.
- Cloud deployment, monitoring and automated CI/CD are outside the present scope.
- The product catalogue and test data are demonstration data.

5.3 Future Scope

- Integrate a secure payment gateway and order-payment verification.
- Deploy the application using cloud hosting and managed MongoDB.
- Add JWT/refresh-token based authentication and role-based access control.
- Add product recommendations, advanced search and personalised offers.
- Introduce automated unit, API and end-to-end testing.
- Add CI/CD pipelines, monitoring, logging and performance optimisation.

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[2] Node.js Documentation, “Node.js JavaScript Runtime.”

[3] Express.js Documentation, “Fast, unopinionated, minimalist web framework for Node.js.”

[4] MongoDB Documentation, “MongoDB Manual and Developer Documentation.”

[5] MDN Web Docs, “HTML, CSS and JavaScript Web Development Documentation.”

[6] Axios Documentation, “Promise-based HTTP client for the browser and Node.js.”

[7] Ice Hut Technologies Pvt. Ltd., Internship/Training Certificate and training documents supplied for the preparation of this report.

ANNEXURE

Training Details

Organisation: Ice Hut Technologies Pvt. Ltd.
Domain: MERN Stack Development
Training Period: 16 May 2026 – 30 June 2026
Mode: Offline
Guide: Mr. Gajendra Badra
Student: Anshu Verma
Roll No.: 24ESKCA025

The internship certificate supplied with the training documents is included in this report as a separate certificate page.